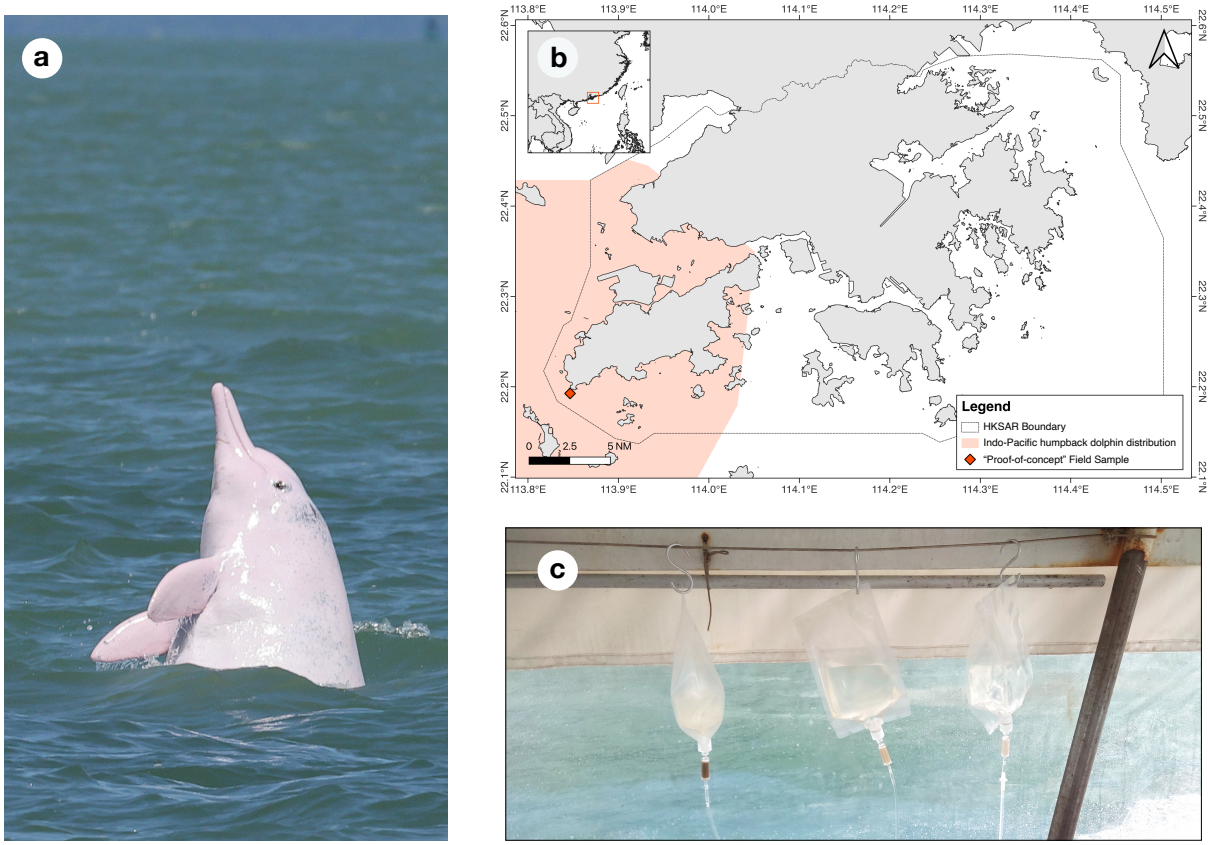


Figure 1 | Indo-Pacific humpback dolphin in Hong Kong waters (**a**) Indo-Pacific humpback dolphin (*Sousa chinensis*), or Chinese white dolphin, in Hong Kong waters (Photo by E. Matrai). (**b**) The distribution of *S. chinensis* around the Lantau island in Hong Kong waters. The red diamond marks the location where we sighted *S. chinensis* and collected the "field positive control" samples for qPCR analysis. (**c**) Gravity filtration of seawater on a boat during the field survey (Photo taken by M. Ushio). The photos are published under the CC BY 4.0 license.

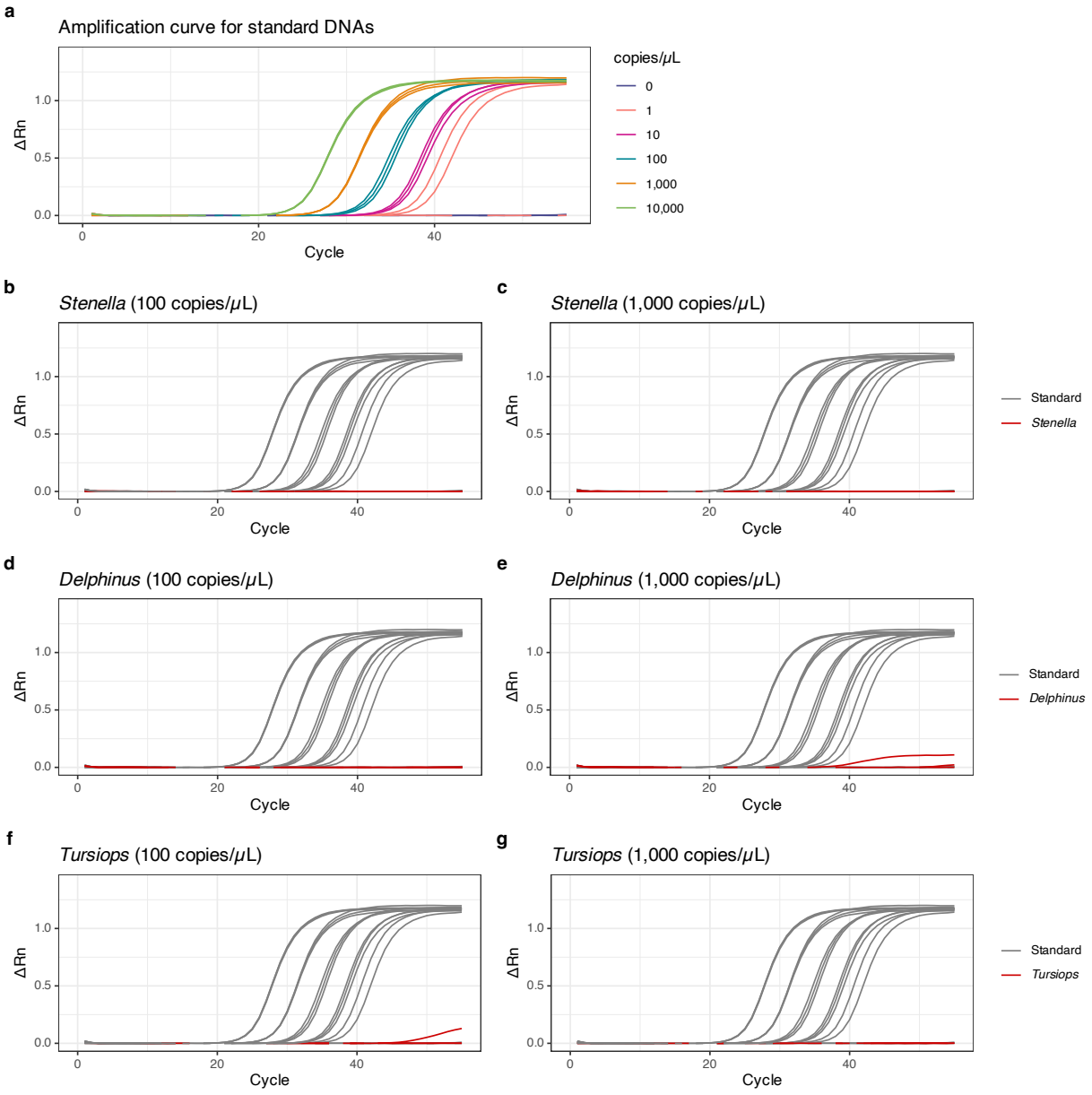


Figure 3 | Specificity tests of the *Sousa chinensis* qPCR assay against three sympatric Delphinidae species. (**a**) Amplification curves of *S. chinensis* standard DNA across a dilution series (0 to 10,000 copies/μL). Different colors indicate different template concentrations. (**b–g**) Amplification curves of synthesized target-region (Cytb) DNA from three non-target, but potentially co-occurring cetacean species (red curves) tested at 100 copies/μL (**b**, **d**, **f**; slightly exceeding the maximum concentration observed in our field positive controls, ~60 copies/μL) and 1,000 copies/μL (**c**, **e**, **g**), plotted alongside the *S. chinensis* standard DNA curves (gray curves, 0 to 10,000 copies/μL): (**b**, **c**) *Stenella coeruleoalba*, (**d**, **e**) *Delphinus capensis* and (**f**, **g**) *Tursiops truncatus*.

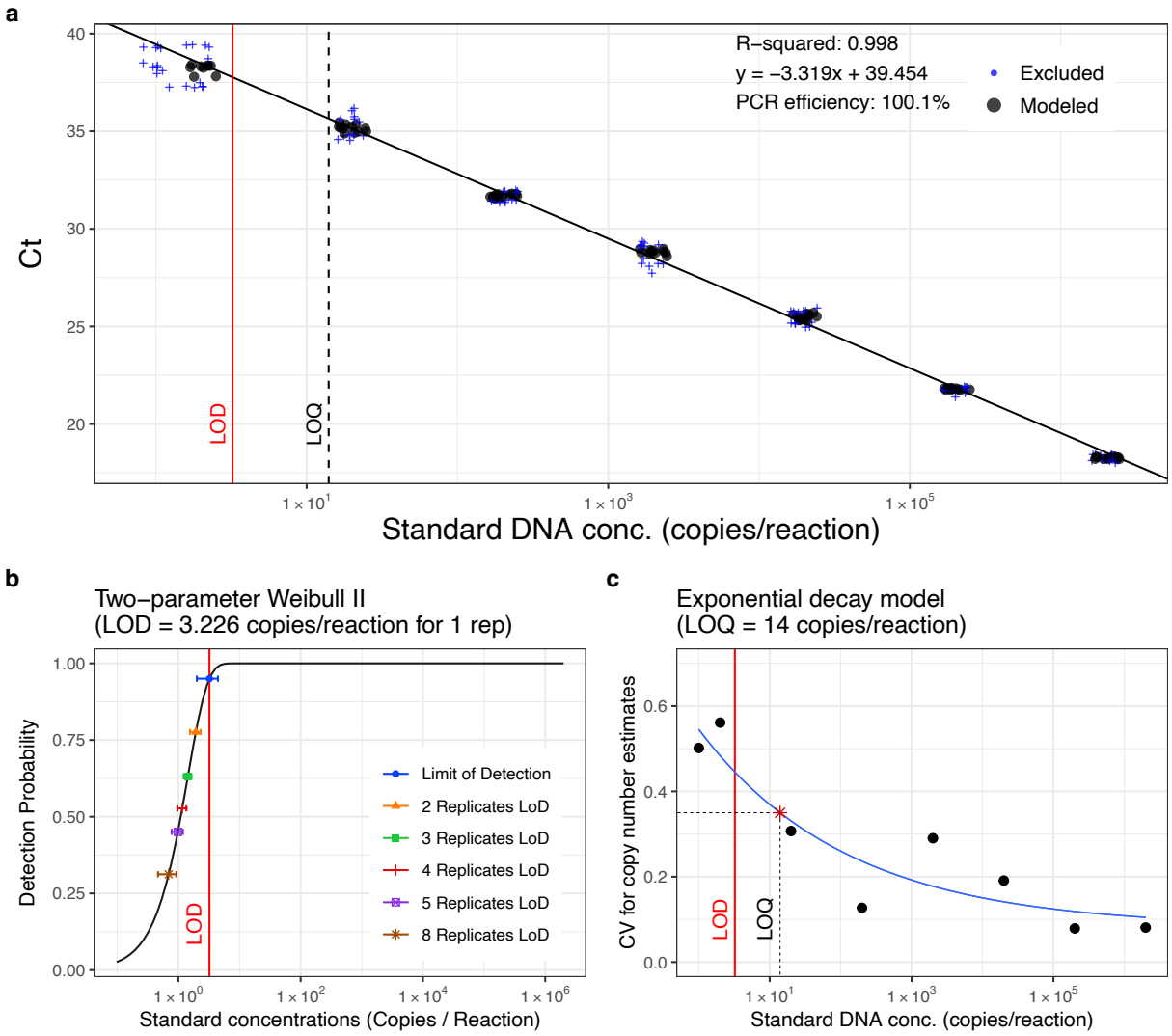


Figure 2 | Standard curve, the limit of detection (LOD) and limit of quantification (LOQ) of the primers and probe specific to *Sousa chinensis*. (**a**) Standard curve generated from 24 technical replicates across eight standard DNA concentrations (1 to 2.0 $\times$ 10^6 copies/reaction). The x-axis represents copy numbers of the *Cytb* target region per qPCR reaction, and the y-axis represents the C_t value. Following Klymus et al. (2020), filled black points indicate data included in the linear regression (the middle two quartiles, i.e., 25th–75th percentiles, of standards with $\geq$ 50% detection rates), whereas blue "+" symbols indicate data excluded from the regression (points outside the middle two quartiles or standards with $<$ 50% detection rates, i.e., 1 copy/reaction). The red solid vertical line and black dashed vertical line indicate the modeled LOD (3.22 copies/reaction) and modeled LOQ (14 copies/reaction), respectively. (**b**) Determination of the LOD. Open circles indicate observed detection rates at each standard concentration. The two-parameter Weibull type II model (`W2.2()` in R) was selected as the best-fitting model. Colored symbols with horizontal error bars indicate effective LODs (with 95% confidence intervals) for different numbers of technical replicates (1 to 8 replicates). (**c**) Determination of the LOQ based on the exponential decay model. The x-axis represents copy numbers and the y-axis represents the coefficient of variation (CV) for copy number estimates calculated following Forootan et al. (2017) and Klymus et al. (2020). The blue curve indicates the exponential decay model (see Supplementary Fig. S2 and Table Sx for comparisons of all candidate models). The red vertical line indicates the modeled LOD, and the black "*" symbol indicates the 35% CV precision threshold, at which the copy number is defined as the modeled LOQ (14 copies/reaction, equivalent to 7.0 copies/μL).

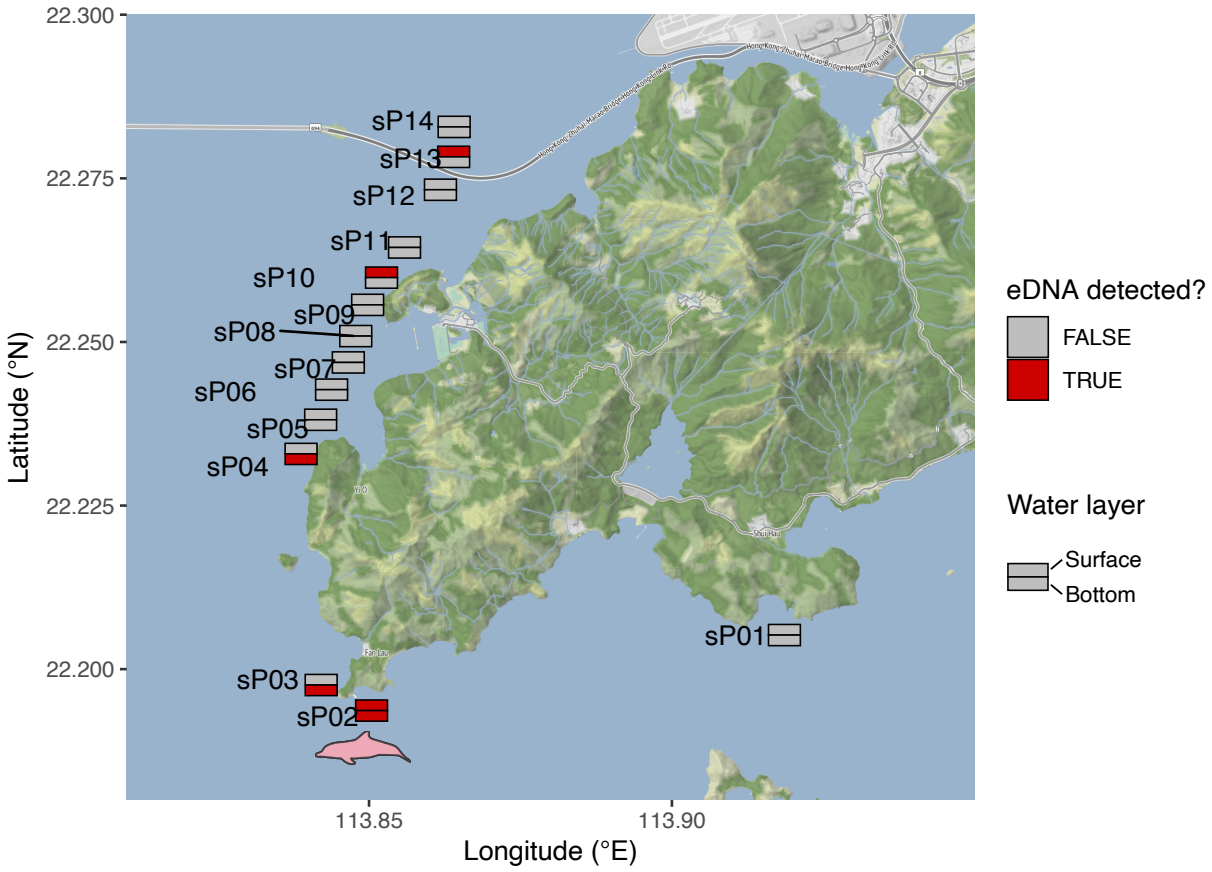


Figure 4 | Detection of *Sousa chinensis* eDNA from field seawater samples using qPCR. qPCR-based detection of *S. chinensis* eDNA from surface and bottom water samples taken in July 2024. Labels indicate the sample collection locations (see the metadata in the supporting information). Red and gray color indicate positive and negative detection, respectively. Upper and lower rectangles indicate the result for surface and bottom water layers, respectively. *S. chinensis* silhouette indicates *S. chinensis* sighting during the field survey (silhouette credit, Chris huh, CC BY-SA, https://creativecommons.org/licenses/by-sa/3.0/). Map data by OpenMapTiles and OpenStreetMap contributors, under ODbL. Maps were generated using the `ggmap` package in R.